

Calculations for network paper

(Dated: June 1, 2026)

AUDIT NOTE: how this annotated copy should be read

This file preserves Luca's original manuscript text and equation order as far as possible. The inserted boxes are the audit layer. Original equations are left in place, even when they are not valid under the stochastic interpretation used here. The correction starts from the model in which a shortcut from u into the absorbing site v removes probability from the transient state space. Under that interpretation the shortcut is a killing channel at u , not a probability-conserving transfer to another transient site.

Source of truth: `source/original_from_outlook_Calculations.tex`. The original source file is archived separately and is not modified by this annotated copy.

Companion full derivation: `Calculations_corrected_complete_v3.tex` contains the expanded 21-page corrected derivation. The present file is the in-place redline map; the companion file is the no-shortcut, no-redline derivation trail behind the blue replacement formulae.

PRELIMINARIES

These are preliminaries identities that can be derived by using the finite series identities in PRX. I have derived the following for $|m| < N$ and $m \neq 0$

$$\frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} = 1 \quad (1)$$

Another identity exploits the fact that

$$\begin{aligned} [1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] &= \frac{1}{2} \left\{ \cos \left[\frac{(|m| - 1)\pi k}{N} \right] - \cos \left[\frac{(|m| + 1)\pi k}{N} \right] \right. \\ &\quad \left. - \frac{1}{2} \left[\cos \left[\frac{(N - |m| + 1)\pi k}{N} \right] + \cos \left[\frac{(N + |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N - |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N + |m| + 1)\pi k}{N} \right] \right] \right\} \end{aligned}$$

for a generic σ we have

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} &= \frac{1}{2} \sum_{k=1}^N \frac{\cos \left[\frac{(|m| - 1)\pi k}{N} \right] - \cos \left[\frac{(|m| + 1)\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} \\ &\quad - \frac{1}{4} \sum_{k=1}^{N-1} \frac{\cos \left[\frac{(N - |m| + 1)\pi k}{N} \right] + \cos \left[\frac{(N + |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N - |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N + |m| + 1)\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} \\ &= \frac{N}{2(\sigma^2 - 1)U_{N-1}(\sigma)} \{T_{N-|m|+1}(\sigma) - T_{N-|m|-1}(\sigma) - T_{|m|-1}(\sigma) + T_{|m|+1}(\sigma)\} \end{aligned} \quad (2)$$

If $\sigma = \cos \left[\frac{2\pi r}{N} \right]$ one needs to use de l'Hospital rule to evaluate the expression above, which gives

$$\begin{aligned} \lim_{\sigma \rightarrow \cos \left[\frac{2\pi r}{N} \right]} \frac{N}{2(\sigma^2 - 1)U_{N-1}(\sigma)} \{T_{N-|m|+1}(\sigma) - T_{N-|m|-1}(\sigma) - T_{|m|-1}(\sigma) + T_{|m|+1}(\sigma)\} \\ = \cos \left[\frac{2\pi r|m|}{N} \right], \end{aligned} \quad (3)$$

in other words we have the general identity for $0 < |m| < N$ and $r = 1, \dots, N - 1$,

$$\frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{\cos \left[\frac{2\pi r}{N} \right] - \cos \left[\frac{\pi k}{N} \right]} = \cos \left[\frac{2\pi r|m|}{N} \right], \quad (4)$$

The third relevant identity for $|m| \leq N$ and integer k is given by

$$\frac{1}{N} \sum_{r=1}^{N-1} \frac{\cos \left[\frac{2\pi r m}{N} \right]}{\cos \left[\frac{(2k-1)\pi}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} = \frac{1}{N \left\{ 1 - \cos \left[\frac{(2k-1)\pi}{N} \right] \right\}} - \frac{\sin \left[\frac{(2k-1)\pi |m|}{N} \right]}{\sin \left[\frac{(2k-1)\pi}{N} \right]} \quad (5)$$

alternatively

$$\frac{1}{N} \sum_{r=1}^{N-1} [1 - (-1)^k] \frac{\cos \left[\frac{2\pi r m}{N} \right]}{\cos \left[\frac{k\pi}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} = [1 - (-1)^k] \left\{ \frac{1}{N \left\{ 1 - \cos \left[\frac{\pi k}{N} \right] \right\}} - \frac{\sin \left[\frac{\pi k |m|}{N} \right]}{\sin \left[\frac{\pi k}{N} \right]} \right\} \quad (6)$$

PROPAGATOR IN RING LATTICE WITH ABSORPTION SITE

The occupation probability for the periodic ring lattice with arbitrary K and an absorption site at $n = m$ can be written via the general relation for any $t \geq 1$

$$W_{n_0}(n, t) = Q_{n_0}^{(k)}(n, t) - \sum_{s=1}^t \mathcal{F}_{n_0 \rightarrow m}(s) Q_m^{(k)}(n, t-s) \quad (7)$$

where $Q_r^{(k)}(n, t)$ is the propagator for the periodic ring lattice in the absence of any absorption site. Note that at $t = 0$, by construction $F_{n_0 \rightarrow m}(0) = 0$ when $n_0 \neq n$, and we have $W_{n_0}(n, 0) = Q_{n_0}^{(k)}(n, 0)$.

More explicit expressions can be obtained for the case $K = 2$, that is the nearest-neighbour lattice walk. We have that

$$Q_{n_0}^{(1)}(n, t) = \frac{1}{N} \sum_{k=0}^{N-1} \cos \left[\frac{2\pi k(n-n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t, \quad (8)$$

and for $t \geq 1$

$$\mathcal{F}_{n_0}(m, t) = \frac{q}{N} \sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{|m-n_0|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^{t-1}. \quad (9)$$

with $\mathcal{F}_{n_0}(m, 0) = 0$.

Given that in Eq. (9) only odd k terms are non-zero, the first-passage probability can be written in compact form as

$$\mathcal{F}_{n_0}(m, t) = \sum_{k=1}^{\frac{N}{2}} f_k(n_0, m) \alpha_k^{t-1}. \quad (10)$$

where

$$f_k(n_0, m) = \frac{2q}{N} \sin \left[\frac{|m-n_0|\pi(2k-1)}{N} \right] \sin \left[\frac{\pi(2k-1)}{N} \right] \quad (11)$$

and

$$\alpha_k = 1 - q + q \cos \left[\frac{\pi(2k-1)}{N} \right]. \quad (12)$$

The generating function of Eq. (10) in compact form can be written as

$$\tilde{\mathcal{F}}_{n_0}(m, z) = z \sum_{k=1}^{N-1} \frac{f_k(n_0, m)}{1 - z\alpha_k} = \frac{U_{N-|m-n_0|-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + U_{|m-n_0|-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}{U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}. \quad (13)$$

where $U_r(y)$ is the Chebyshev polynomial of the second kind.

Also we note that using Eq. (1) one can show that the survival probability

$$\sum_{s=1}^t \mathcal{F}_{n_0}(m, s) = 1 - \frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t,$$

which provides the survival probability

$$R_{n_0}(t) = 1 - \sum_{s=0}^t \mathcal{F}_{n_0}(m, s) = \frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t.$$

Using the above expressions in Eq. (7) and evaluating the time summation, we obtain

$$\begin{aligned} W_{n_0}(n, t) &= \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n - n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\ &+ \frac{1}{N^2} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \\ &+ \frac{1}{N^2} \sum_{k=1}^{N-1} \sum_{r=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \cos \left[\frac{2\pi r(n - m)}{N} \right]}{\cos \left[\frac{\pi k}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} \\ &\times \left\{ \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t - \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \right\} \end{aligned} \quad (14)$$

Using the identities (4) and (6), we reduce the double summation to single summations as follows

$$\begin{aligned} &\sum_{k=1}^{N-1} \sum_{r=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \cos \left[\frac{2\pi r(n - m)}{N} \right]}{\cos \left[\frac{\pi k}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} \left\{ \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t - \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \right\} \\ &= -N \sum_{r=1}^{N-1} \cos \left[\frac{2\pi r|n_0 - m|}{N} \right] \cos \left[\frac{2\pi r(n - m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \\ &- \sum_{k=1}^{N-1} [1 - (-1)^k] \left\{ \frac{\sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} - N \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{|n - m| \pi k}{N} \right] \right\} \\ &\times \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \\ &= -N \sum_{r=1}^{N-1} \cos \left[\frac{2\pi r|n_0 - m|}{N} \right] \cos \left[\frac{2\pi r(n - m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \\ &- \sum_{k=1}^{N-1} [1 - (-1)^k] \left\{ \frac{\sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \right. \\ &\left. - N \sin \left[\frac{|n_0 - m| \pi k}{N} \right] \sin \left[\frac{|n - m| \pi k}{N} \right] \right\} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \end{aligned} \quad (15)$$

and we have

$$\begin{aligned}
W_{n_0}(n, t) &= \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
&\quad - \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-m)}{N} \right] \cos \left[\frac{2\pi k(n_0-m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
&\quad + \frac{1}{N} \sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{\pi k|n_0-m|}{N} \right] \sin \left[\frac{\pi k|n-m|}{N} \right] \left\{ 1 - q + q \cos \left[\frac{\pi k}{N} \right] \right\}^t.
\end{aligned} \tag{16}$$

As $\sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-m)}{N} \right] \cos \left[\frac{2\pi k(n_0-m)}{N} \right] = -1$ and $\sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{\pi k|n_0-m|}{N} \right] \sin \left[\frac{\pi k|n-m|}{N} \right] = 0$, one can show that $Q_{n_0}(n, 0) = \delta_{n, n_0}$. Furthermore when either $n = m$ or $n_0 = m$, one has that $Q_{n_0}(n, t) = 0$ for all t , and it is also easy to see that $Q_{n_0}(n, t \rightarrow \infty) = 0$, which means that all random walks eventually reach the absorbing site $n = m$.

We can rewrite Eq. (16) as

$$W_{n_0}(n, t) = \sum_{k=1}^{N-1} G_k^{(1)}(n_0, n, m) \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t + G_k^{(2)}(n_0, n, m) \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \tag{17}$$

where m is the absorbing site, and with

$$\begin{aligned}
G_k^{(1)}(n_0, n, m) &= \frac{\cos \left[\frac{2\pi k(n-n_0)}{N} \right] - \cos \left[\frac{2\pi k(n-m)}{N} \right] \cos \left[\frac{2\pi k(n_0-m)}{N} \right]}{N} \\
G_k^{(2)}(n_0, n, m) &= \frac{[1 - (-1)^k] \sin \left[\frac{\pi k|n_0-m|}{N} \right] \sin \left[\frac{\pi k|n-m|}{N} \right]}{N}
\end{aligned} \tag{18}$$

In terms of generating functions one has

$$\widetilde{W}_{n_0}(n, z) = \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]} + \sum_{k=1}^{N-1} \frac{G_k^{(2)}(n_0, n, m)}{1 - z \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]} \tag{19}$$

or the equivalent form

$$\widetilde{W}_{n_0}(n, z) = \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z\gamma_k} + \sum_{k=1}^{\frac{N}{2}} \frac{g_k^{(2)}(n_0, n, m)}{1 - z\alpha_k}, \tag{20}$$

and its time-dependent version

$$W_{n_0}(n, t) = \sum_{k=1}^{N-1} G_k^{(1)}(n_0, n, m) \gamma_k^t + \sum_{k=1}^{\frac{N}{2}} g_k^{(2)}(n_0, n, m) \alpha_k^t, \tag{21}$$

where α_k is defined in Eq. (12),

$$\gamma_k = 1 - q + q \cos \left(\frac{2\pi k}{N} \right), \tag{22}$$

and

$$g_k^{(2)}(n_0, n, m) = \frac{2}{N} \sin \left[\frac{\pi(2k-1)|n_0-m|}{N} \right] \sin \left[\frac{\pi(2k-1)|n-m|}{N} \right]. \tag{23}$$

The associated expressions in terms of Chebyshev polynomials of the first and second kind, respectively, $T_r(y)$ and

$U_r(y)$, are

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]} &= \left\{ 2 T_{N-|n-n_0|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + 2 T_{|n-n_0|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ &- T_{N-|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \\ &- T_{N-[|n_0-m|+|n-m| \pmod{N}]} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|n_0-m|+|n-m| \pmod{N}} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \Big\} \\ &\times \left[2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^{-1}, \end{aligned} \quad (24)$$

and

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{G_k^{(2)}(n_0, n, m)}{1 - z \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]} &= \left\{ T_{N-|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + T_{N-|N-|n_0-m|-|n-m||} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ &- T_{|N-|n_0-m|-|n-m||} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \Big\} \\ &\times \left[2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^{-1}, \end{aligned} \quad (25)$$

When $n = n_0$ Eq. (17) reduces to

$$\begin{aligned} W_n(n, t) &= \frac{1}{N} \sum_{k=1}^{N-1} \left\{ \left(1 - \cos^2 \left[\frac{2\pi k(n-m)}{N} \right] \right) \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \right. \\ &\quad \left. + [1 - (-1)^k] \sin^2 \left[\frac{\pi k|n-m|}{N} \right] \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \right\}, \end{aligned} \quad (26)$$

whose generating function is given by the compact expression

$$\begin{aligned} \widetilde{W}_n(n, z) &= \left\{ 2 T_N \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + T_{N-|N-2|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|N-2|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ &- T_{N-[2|n-m| \pmod{N}]} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{[2|n-m| \pmod{N}]} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \Big\} \\ &\times \left\{ 2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\}^{-1} \end{aligned} \quad (27)$$

Note that using Eqs. (19) with $n = n_0$ and (27) when $|n-m| = N/2$ with N even, one can show setting $z = 1$ the following identities

$$\sum_{k=1}^{N-1} \frac{1 - (-1)^k}{1 - \cos \left(\frac{\pi k}{N} \right)} = \frac{N^2}{2}, \quad \sum_{k=1}^{N-1} \frac{1}{1 - \cos \left(\frac{\pi k}{N} \right)} = \frac{2N^2 + 1}{6}, \quad \sum_{k=1}^{N-1} \frac{(-1)^k}{1 - \cos \left(\frac{\pi k}{N} \right)} = \frac{1 - N^2}{6}.$$

For the special case $|n-m| = N/2$ with N even Eq. (27) becomes

$$\widetilde{W}_n(n, z) = \frac{T_N \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - 1}{zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)} = \frac{2 \left[U_{\frac{N}{2}-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^2}{zq U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)} \quad (28)$$

PROPAGATOR WITH ONE SHORTCUT AND ONE ABSORBING SITE

AUDIT NOTE: model statement required before the defect formula

The following original paragraph begins with a probability-conserving shortcut defect formula. That formula is valid when the shortcut transfers probability from one transient site to another transient site. In the present use case, however, the shortcut endpoint is also the absorbing site. Therefore one must state the transient-state model before applying any rank-one identity.

Let $\lambda = \beta(1 - q)$ and let P_0 be the transition matrix on the nonabsorbing transient sites for the ring with absorbing target v already removed. A shortcut $u \rightarrow v$ replaces a fraction λ of the stay-put mass at u by immediate absorption. Hence, in transient space,

$$P = P_0 - \lambda|u\rangle\langle u|. \quad (29)$$

This sign is the key point. The perturbation is negative because probability leaves the transient set at u .

We start from the general propagator with a shortcut from $n = u$ to site $n = v$ and we have

$$\tilde{S}_{n_0}(n, z) = \tilde{P}_{n_0}(n, z) + \frac{z\beta(1 - q)\tilde{P}_{n_0}(u, z) [\tilde{P}_u(n, z) - \tilde{P}_v(n, z)]}{1 - z\beta(1 - q) [\tilde{P}_u(u, z) - \tilde{P}_v(u, z)]}, \quad (30)$$

where $\tilde{P}_r(n, z)$ is the ring propagator generating function without shortcuts. If the ring also has an absorbing site, the modularity of the defect technique allows to replace the (probability conserving) propagator $\tilde{P}_r(n, z)$ with the propagator $\tilde{W}_r(n, z)$ in Eq. (17)

$$\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) + \frac{z\beta(1 - q)\tilde{W}_{n_0}(u, z) [\tilde{W}_u(n, z) - \tilde{W}_v(n, z)]}{1 - z\beta(1 - q) [\tilde{W}_u(u, z) - \tilde{W}_v(u, z)]}. \quad (31)$$

When the absorbing site coincides with the shortcut site, i.e. $m = v$, $\tilde{W}_v(n, z) = 0$ and Eq. (31) simplifies further to

$$\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) + \frac{z\beta(1 - q)\tilde{W}_{n_0}(u, z)\tilde{W}_u(n, z)}{1 - z\beta(1 - q)\tilde{W}_u(u, z)}. \quad (32)$$

ERROR and CORRECTION: shortcut to the absorbing target is killing

ERROR. Eq. (32) is not valid for the stochastic interpretation in which the shortcut endpoint v is absorbing. The original formula has a positive rank-one correction and the denominator $1 - z\beta(1 - q)\tilde{W}_u(u, z)$. This would increase transient occupation, which is the opposite of a shortcut that sends probability to an absorbing target.

REDLINE. The core local edit in Eq. (32) is

$$+ \frac{z\beta(1 - q)\tilde{W}_{n_0}(u, z)\tilde{W}_u(n, z)}{1 - z\beta(1 - q)\tilde{W}_u(u, z)} \implies - \frac{z\beta(1 - q)\tilde{W}_{n_0}(u, z)\tilde{W}_u(n, z)}{1 + z\beta(1 - q)\tilde{W}_u(u, z)}. \quad (33)$$

CORRECTION. Starting from Eq. (29),

$$\begin{aligned} (I - zP)^{-1} &= (I - zP_0 + z\lambda|u\rangle\langle u|)^{-1} \\ &= (I - zP_0)^{-1} - \frac{z\lambda(I - zP_0)^{-1}|u\rangle\langle u|(I - zP_0)^{-1}}{1 + z\lambda\langle u|(I - zP_0)^{-1}|u\rangle}. \end{aligned} \quad (34)$$

Therefore the corrected transient propagator is

$$\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) - \frac{z\beta(1 - q)\tilde{W}_{n_0}(u, z)\tilde{W}_u(n, z)}{1 + z\beta(1 - q)\tilde{W}_u(u, z)}. \quad (35)$$

WHY. The plus sign in the denominator and the minus sign in the correction are forced by the Sherman–Morrison identity applied to $P = P_0 - \lambda|u\rangle\langle u|$.

After summing Eq. (31) over n to obtain the survival probability $\tilde{S}_{n_0}(z) = [1 - \tilde{F}_{n_0}(v, z)] / (1 - z)$, one expresses the generating function of the first-passage probability to site v as

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) - \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)]}{1 - z\beta(1-q)\tilde{W}_u(u, z)}, \quad (36)$$

or more explicitly as

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) - qz\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)] \tilde{H}_{N,|u-v|}(z), \quad (37)$$

ERROR and CORRECTION: first-passage sign

ERROR. Eqs. (36)–(37) inherit the wrong sign from Eq. (32). They make the shortcut contribution negative. For $0 \leq z < 1$, the factors $\tilde{W}_{n_0}(u, z)$ and $1 - \tilde{\mathcal{F}}_u(v, z)$ have non-negative coefficients, so a shortcut into the target should increase the hitting generating function.

REDLINE. The original first-passage correction should be crossed out and replaced as

$$\begin{aligned} \tilde{\mathcal{F}}_{n_0}(v, z) - \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)]}{1 - z\beta(1-q)\tilde{W}_u(u, z)} \\ \Rightarrow \tilde{\mathcal{F}}_{n_0}(v, z) + \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)]}{1 + z\beta(1-q)\tilde{W}_u(u, z)}. \end{aligned} \quad (38)$$

CORRECTION. Summing Eq. (35) over all transient sites gives

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)]}{1 + z\beta(1-q)\tilde{W}_u(u, z)}. \quad (39)$$

Equivalently,

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + qz\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)] \tilde{H}_{N,|u-v|}^{\text{corr}}(z), \quad (40)$$

where the corrected auxiliary factor is defined below.

with $\tilde{W}_{n_0}(u, z)$ and $\tilde{\mathcal{F}}_r(m, z)$ defined, respectively, in Eqs. (19) and (13), and

$$\tilde{H}_{N,|u-v|}(z) = \frac{\beta(1-q)}{q} \frac{1}{1 - z\beta(1-q)\tilde{W}_u(u, z)}. \quad (41)$$

CORRECTION: auxiliary denominator

CORRECTION. The corrected auxiliary factor corresponding to Eq. (40) is

$$\frac{1}{1 - z\beta(1-q)\tilde{W}_u(u, z)} \Rightarrow \frac{1}{1 + z\beta(1-q)\tilde{W}_u(u, z)}. \quad (42)$$

Thus

$$\tilde{H}_{N,|u-v|}^{\text{corr}}(z) = \frac{\beta(1-q)}{q} \frac{1}{1 + z\beta(1-q)\tilde{W}_u(u, z)}. \quad (43)$$

This is the denominator change that propagates through the rest of the shortcut analysis.

We now consider the symmetric case $|u - v| = N/2$ with N even, and use $\widetilde{\mathcal{W}}_u(u, z)$ from Eq. (28) in $\widetilde{W}_u(u, z)$, that allows to simplify Eq. (41) to

$$\widetilde{H}_{N,N/2}(z) = \frac{1}{\frac{q}{\beta(1-q)} - \phi(z)}, \quad (44)$$

ERROR and CORRECTION: symmetric-case pole geometry

ERROR. Eq. (44) uses $q/[\beta(1-q)] - \phi(z)$ because it came from the wrong-sign transient resolvent. Consequently the later figure analysis based on a positive horizontal line intersecting $\phi(1/s)$ is not the pole geometry of the corrected stochastic model.

REDLINE. The denominator-level replacement is

$$\frac{1}{\frac{q}{\beta(1-q)} - \phi(z)} \Rightarrow \frac{1}{\frac{q}{\beta(1-q)} + \phi(z)}. \quad (45)$$

CORRECTION. In the symmetric case the auxiliary factor is

$$\widetilde{H}_{N,N/2}^{\text{corr}}(z) = \frac{1}{\frac{q}{\beta(1-q)} + \phi(z)}. \quad (46)$$

Equivalently, if the same plotted curve $\phi(1/s)$ is retained, the horizontal line is $-q/[\beta(1-q)]$, not $+q/[\beta(1-q)]$.

CORRECTED CONTINUATION. Let $N = 2L$, $v = 0$, $u = L$, and let ρ be the starting distance from v along either half-ring. Define

$$a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right). \quad (47)$$

The corrected first-passage generating function is

$$\widetilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}, \quad U_{-1} = 0. \quad (48)$$

This formula replaces the downstream pole analysis based on Eq. (44).

where

$$\phi(z) = \frac{2U_{N/2-1}^2 \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}{U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}. \quad (49)$$

whose time-dependence is $H_{N,N/2}(0) = \beta(1-q)/q$ and for $t \geq 1$ is given by the complex integral (after the change of integration variable $z = 1/s$ and the region of integration becoming $|s| \leq 1$)

$$H_{N,N/2}(t) = \frac{1}{2\pi i} \oint_{|s| \leq 1} \frac{s^{t-1}}{\frac{q}{\beta(1-q)} - \phi\left(\frac{1}{s}\right)}. \quad (50)$$

The denominator in the integrand of Eq. (50) has M real zeros s_k ($k = 1, 2, \dots, M$) and are shown graphically in Fig. 1 (see the figure caption for an explanation of when $M = N/2$ and $M = N/2 - 1$).

AUDIT NOTE: original Fig. 1 is downstream of the wrong sign

The original caption is preserved above. Under the corrected model, the intersection picture must be redrawn with the horizontal line $+q/[\beta(1-q)]$ replaced by $-q/[\beta(1-q)]$. Also note the typographical error in the original caption: it says “~~given the requirement $q > 1$~~ ”, but the stochastic model requires $0 < q < 1$ and a nonzero shortcut probability proportional to $1 - q$ requires $q < 1$.

Original figure file `phifunction.pdf` was not included with the Outlook TeX attachment. The annotated copy keeps the original figure call as a portable placeholder.

FIG. 1. Graphical representation of the poles of Eq. (44) for the case $N = 10$, $q = 2/3$ and $\beta = 2/7$ which are the values of s in the range $-1 \leq s \leq 1$ where the horizontal dotted line $\frac{q}{\beta(1-q)}$ intercepts the function $\phi(s^{-1})$. Depending on N , β and q (N even, $0 < \beta \leq 1$ and $0 < q < 1$) one may have $M = \frac{N}{2} - 1$ poles, when $\frac{q}{\beta(1-q)} < \frac{N}{2}$, or $M = \frac{N}{2}$ poles, when $\frac{q}{\beta(1-q)} \geq \frac{N}{2}$, which is the case displayed above. The asymptotes of the $\phi(s^{-1})$ function are given from right to left by, respectively, α_k in Eq. (12) with $k = 1, 2, \dots, \frac{N}{2}$, and they are interlaced with the poles s_k with $k = 1, \dots, M$. We follow the same principle and order from right to left the poles s_k with increasing k , thus with s_1 representing the largest positive pole. With $q = 1$ the curves are symmetric around the origin, but given the requirement $q > 1$ to have a non-zero probability to move from site u to site v , the function $\phi(s^{-1})$ is never symmetric around $s = 0$, which implies that in magnitude one has $|s_{\frac{N}{2}}| < s_1$ for any $q < 1$. The paired tick marks without numerical values on the top and bottom horizontal axes correspond, from right to left, to the values γ_k in Eq. (22) with $k = 1, 2, \dots, \frac{N}{2}$, which are interlaced with the α_k (the values for $N/2 + 1 \leq k \leq N - 1$ repeat themselves).

s_k (real) poles in the integrand with $k = 1, \dots, N/2 - 1$ in the dominator ($-1 < s_k < 1$) can be found numerically as

$$H_{N,N/2}(t) = \sum_{k=1}^M c_k s_k^{t-1} \quad (51)$$

with

$$c_k = \frac{2q}{N} \frac{\left\{ \left(1 + \frac{1}{q} [s_k - 1]\right)^2 - 1 \right\} \left\{ T_{\frac{N}{2}} \left(1 + \frac{1}{q} [s_k - 1]\right) \right\}^2}{\left(1 + \frac{1}{N}\right) \left(1 + \frac{1}{q} [s_k - 1]\right) U_{N-1} \left(1 + \frac{1}{q} [s_k - 1]\right) - T_N \left(1 + \frac{1}{q} [s_k - 1]\right) - U_{N-2} \left(1 + \frac{1}{q} [s_k - 1]\right) - 1}. \quad (52)$$

Analysis of c_k shows that $c_k \geq 0$ for any real s_k and one may have $c_k = 0$ only when $s_k = \alpha_k$. As α_k are the asymptotes of $\phi(1/s)$, then we conclude that $c_k > 0$ for all k .

The time dependence of Eq. (37) for $|u - v| = N/2$ is thus given by the time-convoluted expression for $t \geq 1$

$$F_{n_0}(v, t) = \mathcal{F}_{n_0}(v, t) - q \sum_{t'=0}^{t-1} W_{n_0}(u, t-1-t') \sum_{t''=0}^{t'} [\delta_{t',t''} - \mathcal{F}_u(v, t' - t'')] H_{N,N/2}(t'') \quad (53)$$

and $F_{n_0}(v, 0) = 0$. Using Eqs. (10), (21) and (51), after performing the various time convolutions, one obtains

$$\begin{aligned} F_{n_0}(v, t) = & \sum_{\ell=1}^{\frac{N}{2}} f_{\ell}(n_0, u) \alpha_{\ell}^{t-1} - q \sum_{k=1}^M \left\{ \sum_{r=1}^{N-1} \frac{c_k G_r^{(1)}(n_0, u, v)}{s_k(s_k - \gamma_r)} (s_k^t - \gamma_r^t) + \sum_{\ell=1}^{\frac{N}{2}} \frac{c_k g_{\ell}^{(2)}(n_0, u, v)}{s_k(s_k - \alpha_{\ell})} (s_k^t - \alpha_{\ell}^t) \right\} \\ & + q \sum_{k=1}^M \sum_{r=1}^{N-1} \sum_{\ell=1}^{\frac{N}{2}} \frac{c_k G_r^{(1)}(n_0, u, v) f_{\ell}(u, v)}{s_k - \alpha_{\ell}} \left[\frac{s_k^t - \gamma_r^t}{\alpha_{\ell}(s_k - \gamma_r)} - \frac{\alpha_{\ell}^t - \gamma_r^t}{s_k(\alpha_{\ell} - \gamma_r)} \right] - q \sum_{k=1}^M \sum_{\ell=1}^{\frac{N}{2}} \frac{c_k g_{\ell}^{(2)}(n_0, u, v) f_{\ell}(u, v)}{s_k \alpha_{\ell}(s_k - \alpha_{\ell})} \alpha_{\ell}^t t \\ & + q \underbrace{\sum_{k=1}^M \sum_{r=1}^{\frac{N}{2}} \sum_{\ell=1}^{\frac{N}{2}}}_{\ell \neq r} \frac{c_k g_r^{(2)}(n_0, u, v) f_{\ell}(u, v)}{s_k \alpha_{\ell}(s_k - \alpha_{\ell})} \left[s_k \frac{s_k^t - \gamma_r^t}{s_k - \gamma_r} - \alpha_{\ell} \frac{\alpha_{\ell}^t - \gamma_r^t}{\alpha_{\ell} - \gamma_r} \right]. \quad (54) \end{aligned}$$

ERROR and CORRECTION: downstream convolution should be replaced

ERROR. Eq. (54) is downstream of the wrong $\tilde{H}$ denominator, so it should not be used as the final time-domain answer. There are also local bookkeeping problems: the baseline no-shortcut first-passage term should point to v , not u , and the odd-sector propagator terms involving $g_r^{(2)}$ should be paired with the odd-sector eigenvalue α_r rather than γ_r .

REDLINE. Even before replacing the whole expression, two local edits that a reader should notice are

$$\cancel{f_{\ell}(n_0, u)} \implies f_{\ell}(n_0, v), \quad \cancel{\gamma_r \text{ on an odd-sector } g_r^{(2)} \text{ term}} \implies \alpha_r. \quad (55)$$

CORRECTION. Use the corrected closed form Eq. (48) and expand it by residues. Let

$$D(y) = aT_L(y) + U_{L-1}(y), \quad N_\rho(y) = aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y). \quad (56)$$

If y_j are the roots of D and $s_j = 1 - q + qy_j$, then

$$F_\rho(t) = \sum_{j=1}^L B_{\rho j} s_j^{t-1}, \quad B_{\rho j} = \frac{q N_\rho(y_j)}{D'(y_j)}. \quad (57)$$

This is shorter and keeps the coefficients tied directly to the corrected pole equation.

The exponentially decaying dominant term in the first-passage probability can now be evinced by finding the largest positive coefficient among α_k , γ_k and s_k with $k = 1$ identifying the largest value for each one. As $\alpha_1 > \gamma_1$, it suffices to compare α_1 and s_1 . As graphically in Fig. 1 we have observed that when $q/[\beta(1-q)] \geq N/2$, the largest coefficient is the pole s_1 , whereas it is α_1 if $q/[\beta(1-q)] < N/2$, we are able to identify analytically the necessary condition for the second peak to be present, and that is $\beta \leq \frac{2q}{(1-q)N}$. From the last term in the second line of Eq. (54) one may also notice that when $\beta > \frac{2q}{(1-q)N}$ the long time dependence is dominated by $\alpha_1^t t$, which indicates a logarithmic correction that slows the exponential decay. In other words the introduction of the shortcut between u and v changes the long time dependence of the first-passage probability from α_1^t to either $\alpha_1^t t$ when $\beta \geq \frac{2q}{(1-q)N}$ or to s_1^t when $\beta < \frac{2q}{(1-q)N}$. While the case $\beta = \frac{2q}{(1-q)N}$ corresponds to the transition from $\alpha_1^t t$ to s_1^t , $s_1 = 1$ and thus it does not affect the long time dependence.

ERROR and CORRECTED CONCLUSION: tail asymptotics vs finite-time bimodality

ERROR. The preceding original paragraph is the main conclusion that changes. The claimed threshold

$$\beta \leq \frac{2q}{(1-q)N}$$

comes from the wrong-sign denominator in Eq. (44). It is not a valid analytical threshold for the corrected stochastic shortcut-to-absorbing target model. The asserted $t\alpha_1^t$ term is a repeated-pole artefact of the uncorrected pole collision.

REDLINE. The conclusion-level replacement is

$$\beta \leq \frac{2q}{(1-q)N} \text{ as corrected condition} \implies \text{no such tail-dominance threshold follows from the corrected model}, \quad (58)$$

and

$$t\alpha_1^t \text{ or } s_1^t \implies B_{\rho 1} s_1^{t-1} \text{ for the corrected right tail.} \quad (59)$$

CORRECTED CONTINUATION. The corrected pole equation is

$$aT_L(y_s) + U_{L-1}(y_s) = 0, \quad y_s = 1 + \frac{s-1}{q}. \quad (60)$$

Equivalently, writing $y_s = \cos \theta$,

$$\tan(L\theta) = -a \sin \theta. \quad (61)$$

The largest positive root satisfies

$$\gamma_1 < s_1 < \alpha_1, \quad \gamma_1 = 1 - q + q \cos \frac{2\pi}{N}, \quad \alpha_1 = 1 - q + q \cos \frac{\pi}{N}. \quad (62)$$

Therefore the corrected right tail is

$$F_\rho(t) \sim B_{\rho 1} s_1^{t-1}, \quad t \rightarrow \infty, \quad (63)$$

not generically $t\alpha_1^t$.

ADDED DECAY DEPENDENCE. For weak shortcut strength, perturbing the corrected pole equation gives

$$s_1 = \alpha_1 - \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2), \quad (64)$$

and hence

$$1 - s_1 = 1 - \alpha_1 + \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2). \quad (65)$$

For large N this becomes

$$1 - s_1 \sim \frac{q\pi^2}{2N^2} + \frac{2\beta(1-q)}{N}. \quad (66)$$

FINITE-TIME BIMODALITY. Tail dominance is not the same question as whether the finite-time first-passage curve has two visually resolved peaks. The corrected formula above determines the asymptotic right tail. A finite-time double peak should be checked from the corrected finite-time series, with an explicit peak/valley criterion.

CORRECTED REPLACEMENT PARAGRAPH FOR THE AUTHOR

The shortcut-to-target calculation should be rewritten as follows. In the transient state space, a shortcut from u to the absorbing site v is an additional killing channel at u , so the transient matrix is $P = P_0 - \beta(1-q)|u\rangle\langle u|$. The rank-one resolvent identity then gives the denominator $1 + z\beta(1-q)\widetilde{W}_u(u, z)$, not $1 - z\beta(1-q)\widetilde{W}_u(u, z)$. Consequently the corrected symmetric first-passage generating function is

$$\widetilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}, \quad a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right).$$

The corrected poles solve $aT_L(y_s) + U_{L-1}(y_s) = 0$, or $\tan(L\theta) = -a \sin \theta$ when $y_s = \cos \theta$. The largest positive pole satisfies $\gamma_1 < s_1 < \alpha_1$, so the right tail is $F_\rho(t) \sim B_{\rho 1} s_1^{t-1}$ rather than a generic $t\alpha_1^t$ term. The original condition $\beta \leq 2q/[(1-q)N]$ is therefore not a corrected analytical condition for bimodality; finite-time bimodality must be assessed from the corrected finite-time series using an explicit peak/valley diagnostic.